

Q1: 24 Feb (Shift 1) - Numerical

If the least and the largest real values of α , for which the equation $z + \alpha|z - 1| + 2i = 0$ ($z \in \mathbb{C}$ and $i = \sqrt{-1}$) has a solution, are p and q respectively; then $4(p^2 + q^2)$ is equal to

Q2: 24 Feb (Shift 2) - Numerical

Let $i = \sqrt{-1}$. If $\frac{(-1+i\sqrt{3})^{21}}{(1-i)^{24}} + \frac{(1+i\sqrt{3})^{21}}{(1+i)^{24}} = k$, and $n = [|k|]$ be the greatest integral part of $|k|$. Then $\sum_{j=0}^{n+5} (j+5)^2 - \sum_{j=0}^{n+5} (j+5)$ is equal to

Q3: 25 Feb (Shift 1) - Single Correct

Let the lines $(2-i)z = (2+i)\bar{z}$ and $(2+i)z + (i-2)\bar{z} - 4i = 0$, (here $i^2 = -1$) be normal to a circle C . If the line $iz + \bar{z} + 1 + i = 0$ is tangent to this circle C , then its radius is :

(1) $\frac{3}{\sqrt{2}}$

(2) $3\sqrt{2}$

(3) $\frac{3}{2\sqrt{2}}$

(4) $\frac{1}{2\sqrt{2}}$

Q4: 26 Feb (Shift 1) - Numerical

The sum of 162^{th} power of the roots of the equation $x^3 - 2x^2 + 2x - 1 = 0$ is

Q5: 26 Feb (Shift 2) - Numerical

Let z be those complex number which satisfy $|z+5| \leq 4$ and $z(1+i) + \bar{z}(1-i) \geq -10$, $i = \sqrt{-1}$. If the maximum value of $|z+1|^2$ is $\alpha + \beta\sqrt{2}$, then the value of $(\alpha + \beta)$ is

Answer Key**Q1 (10)****Q2 (310)****Q3 (3)****Q4 (3)****Q5 (48)**

Q1 (10)

$$x + iy + \alpha\sqrt{(x-1)^2 + y^2} + 2i = 0$$

$$\therefore y + 2 = 0 \text{ and } x + \alpha\sqrt{(x-1)^2 + y^2} = 0$$

$$y = -2 \text{ and } x^2 = \alpha^2 (x^2 - 2x + 1 + 4)$$

$$\alpha^2 = \frac{x^2}{x^2 - 2x + 5} \Rightarrow x^2 (\alpha^2 - 1) - 2x\alpha^2 + 5\alpha^2 = 0$$

$$x \in R \Rightarrow D \geq 0$$

$$\alpha\alpha^4 - 4(\alpha^2 - 1)5\alpha^2 \geq 0$$

$$\alpha^2 [4\alpha^2 - 2\alpha^2 + 20] \geq 0$$

$$\alpha^2 [-16\alpha^2 + 20] \geq 0$$

$$\alpha^2 \left[\alpha^2 - \frac{5}{4} \right] \leq 0$$

$$0 \leq \alpha^2 \leq \frac{5}{4}$$

$$\therefore \alpha^2 \in \left[0, \frac{5}{4} \right]$$

$$\therefore \alpha \in \left[-\frac{\sqrt{5}}{2}, \frac{\sqrt{5}}{2} \right]$$

$$\text{then } 4[(q)^2 + (p)^2] = 4\left[\frac{5}{4} + \frac{5}{4}\right] = 10$$

Q2 (310)

$$\frac{\left(2e^{i\frac{2\pi}{3}}\right)^{21}}{\left(\sqrt{2}e^{-i\frac{\pi}{4}}\right)^{24}} + \frac{\left(2e^{i\frac{\pi}{3}}\right)^{21}}{\left(\sqrt{2}e^{i\frac{\pi}{4}}\right)^{24}}$$

$$\Rightarrow \frac{2^{21} \cdot e^{i14\pi}}{2^{12} \cdot e^{-i6\pi}} + \frac{2^{21} (e^{i7\pi})}{2^{12} (e^{i6\pi})}$$

$$\Rightarrow 2^9 e^{i(20\pi)} + 2^9 e^{i\pi}$$

$$\Rightarrow 2^9 + 2^9(-1) = 0$$

$$n = 0$$

$$\sum_{j=0}^5 (j+5)^2 - \sum_{j=0}^5 (j+5)$$

$$\Rightarrow [5^2 + 6^2 + 7^2 + 8^2 + 9^2 + 10^2] - [5 + 6 + 7 + 8 + 9 + 10]$$

$$\Rightarrow [(1^2 + 2^2 + \dots + 10^2) - (1^2 + 2^2 + 3^2 + 4^2)] - [(1 + 2 + 3 + \dots + 10) - (1 + 2 + 3 + 4)]$$

$$\Rightarrow (385 - 30) - [55 - 10]$$

$$\Rightarrow 355 - 45 \Rightarrow 310 \text{ ans.}$$

Q3 (3)

$$(2 - i)z = (2 + i)\bar{z}$$

$$\Rightarrow (2 - i)(x + iy) = (2 + i)(x - iy)$$

$$\Rightarrow 2x - ix + 2iy + y = 2x + ix - 2 - iy + y$$

$$\Rightarrow 2ix - 4iy = 0$$

$$L_1 : x - 2y = 0$$

$$\Rightarrow (2 + i)z + (i - 2)\bar{z} - 4i = 0$$

$$\Rightarrow (2 + i)(x + iy) + (i - 2)(x - iy) - 4i = 0$$

$$\Rightarrow 2x + ix + 2iy - y + ix - 2x + y + 2iy - 4i = 0$$

$$\Rightarrow 2ix + 4iy - 4i = 0$$

$$L_2 : x + 2y - 2 = 0$$

$$\text{Solve } L_1 \text{ and } L_2 \quad 4y = 2, \quad y = \frac{1}{2}$$

$$\therefore x = 1$$

$$\text{Centre } \left(1, \frac{1}{2}\right)$$

$$L_3 : iz + \bar{z} + 1 + i = 0$$

$$\Rightarrow i(x + iy) + x - iy + 1 + i = 0$$

$$\Rightarrow ix - y + x - iy + 1 + i = 0$$

$$\Rightarrow (x - y + 1) + i(x - y + 1) = 0$$

$$\text{Radius} = \text{distance from } \left(1, \frac{1}{2}\right) \text{ to } x - y + 1 = 0$$

$$r = \frac{1 - \frac{1}{2} + 1}{\sqrt{2}}$$

$$r = \frac{3}{2\sqrt{2}}$$

Q4 (3)

$$\text{Let roots of } x^3 - 2x^2 + 2x - 1 = 0 \text{ are } \alpha, \beta, \gamma$$

$$(x - 1)(x^2 - x + 1) = 0$$

$$\text{Now } \alpha^{162} + \beta^{162} + \gamma^{162}$$

$$= 1 + \omega^{162} + (\omega^2)^{162}$$

$$= 1 + (\omega^3)^{54} + (\omega^3)^{108}$$

$$= 3$$

Q5 (48)

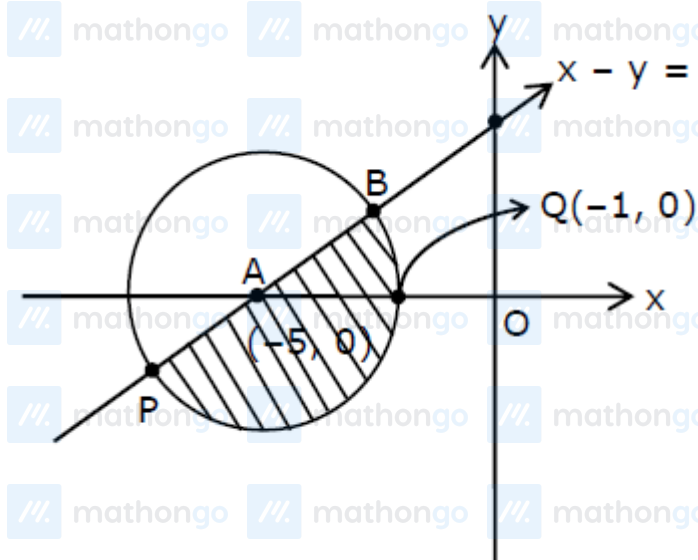
Given, $|z + 5| \leq 4$

$$\Rightarrow (x + 5)^2 + y^2 \leq 16 \quad \dots (1)$$

Also, $z(1 + i) + \bar{z}(1 - i) \geq -10$.

$$\Rightarrow x - y \geq -5 \quad \dots (2)$$

From (1) and (2) Locus of z is the shaded region in the diagram.



$|z + 1|$ represents distance of ' z ' from $Q(-1, 0)$. Clearly ' P ' is the required position of ' z ' when $|z + 1|$ is maximum. $\therefore P \equiv (-5 - 2\sqrt{2}, -2\sqrt{2})$

$$\therefore (PQ)^2|_{\max} = 32 + 16\sqrt{2}$$

$$\Rightarrow \alpha = 32$$

$$\Rightarrow \beta = 16$$

Thus, $\alpha + \beta = 48$